

$$q_{\pi}(s, a)$$

$$V_{\pi}(s)$$



Action-Value in Terms of State-Value?

$$q_{\pi}(s, a) = E_{\pi} [R_{t+1} + \gamma R_{t+2} + \dots \mid S_t = s, A_t = a]$$

$$V_{\pi}(s) = E_{\pi} [R_{t+1} + \gamma R_{t+2} + \dots \mid S_t = s]$$

$$\rightarrow V_{\pi}(s') = E_{\pi} [R_{t+2} + \gamma R_{t+3} + \gamma^2 R_{t+4} + \dots \mid S_{t+1} = s']$$

$$\rightarrow = E_{\pi} [R_{t+1} \mid S_t = s] + \gamma E_{\pi} [R_{t+2} + \gamma R_{t+3} + \dots \mid S_t = s]$$

$$= E_{\pi} [R_{t+1} \mid S_t = s] + \gamma E_{\pi} \left[\underbrace{E_{\pi} [R_{t+2} + \gamma R_{t+3} + \dots \mid S_t = s, S_{t+1} = s']}_{= E_{\pi} [R_{t+2} + \gamma R_{t+3} + \dots \mid S_{t+1} = s']} \mid S_t = s \right]$$

$$\begin{aligned}
 & E[g(X, Y) | X=x] \\
 &= E[E[g(X, Y) | X=x, Y=y] | X=x]
 \end{aligned}
 \left\{
 \begin{aligned}
 & E[XY] \\
 &= E[E[XY | Y]] \\
 &= E[Y E[X | Y]] \\
 &= \int y \left[\int x f_{X|Y}(x|y) dx \right] f_Y(y) dy \\
 &= \int xy f_{X,Y}(x,y) dx dy.
 \end{aligned}
 \right.
 \begin{array}{l}
 \text{Solving } E[XY] \\
 \text{using iterated} \\
 \text{expectation.}
 \end{array}$$

$$q_\pi(s, a) = E_\pi[R_{t+1} + \gamma R_{t+2} + \dots | S_t = s, A_t = a]$$

$$V_\pi(s) = E_\pi[R_{t+1} + \gamma R_{t+2} + \dots | S_t = s]$$

$$V_\pi(s') = E_\pi[R_{t+2} + \gamma R_{t+3} + \gamma^2 R_{t+4} + \dots | S_{t+1} = s']$$

$$= E_\pi[R_{t+1} | S_t = s] + \gamma E_\pi[R_{t+2} + \gamma R_{t+3} + \dots | S_t = s]$$

$$= E_\pi[R_{t+1} | S_t = s] + \gamma E_\pi \left[\underbrace{E_\pi[R_{t+2} + \gamma R_{t+3} + \dots | S_t = s, S_{t+1} = s']}_{\substack{\downarrow \\ = E_\pi[R_{t+2} + \gamma R_{t+3} + \dots | S_{t+1} = s'] = V_\pi(s')}} | S_t = s \right]$$

$$= E_\pi[R_{t+1} | S_t = s] + \gamma E_\pi[V_\pi(S_{t+1}) | S_t = s]$$

$$V_\pi(s) = E_\pi[R_{t+1} + \gamma V_\pi(S_{t+1}) | S_t = s]$$

$$q_\pi(s, a) = E_\pi \left[\underbrace{R_{t+1}}_{\downarrow} + \gamma V_\pi(S_{t+1}) \mid S_t = s, A_t = a \right]$$

$$PMF(s) \quad P[S_{t+1} = s', R_{t+1} = r \mid S_t = s, A_t = a] \text{ for every } (s, a)$$

↳ MDP

Bellman Equations for State-Value, Action-Value

- Rewrite the state-value function in terms of the function calculated at the *next* state

$v_{\pi}(S_t)$ in terms of $v_{\pi}(S_{t+1})$

$$v_{\pi}(s) = E_{\pi} \left[\underbrace{R_{t+1} + \gamma v_{\pi}(S_{t+1})}_{\text{next state value}} \mid S_t = s \right]$$

Bellman Equation Using the MDP

$$v_\pi(s) = E[\underbrace{R_{t+1}}_{\text{reward}} + \underbrace{\gamma v_\pi(S_{t+1})}_{\text{discounted future value}} | \underbrace{S_t = s}_{\text{current state}}]$$

$$q_\pi(s, a) = \underset{\text{blue}}{E}[R_{t+1} + \gamma v_\pi(S_{t+1}) | S_t = s, A_t = a]$$

Expand the E operator:

$$V_{\pi}(s) = \sum_{s'} \underbrace{\sum_{r} (r + \gamma V_{\pi}(s'))}_{\Rightarrow \sum_a p[R_{t+1}=r, S_{t+1}=s', A_t=a | S_t=s]} p[R_{t+1}=r, S_{t+1}=s' | S_t=s]$$

Expand the E operator:

$$V_{\pi}(s) = \sum_{s'} \sum_{r} \underbrace{(r + \gamma V_{\pi}(s'))}_{\text{}} \underbrace{P[R_{t+1}=r, S_{t+1}=s' | S_t=s]}_{\text{}} \\ \Rightarrow \sum_a P[R_{t+1}=r, S_{t+1}=s', A_t=a | S_t=s]$$

$$= \sum_a \sum_{s'} \sum_{r} (r + \gamma V_{\pi}(s)) P[R_{t+1}=r, S_{t+1}=s' | S_t=s, A_t=a] \\ \underbrace{P[A_t=a | S_t=s]}_{\downarrow \pi(a|s)}$$

Bellman Equation Using the MDP

- System of linear equations; one for each state s

$$\begin{aligned} v_{\pi}(s) &= E[R_{t+1} + \gamma v_{\pi}(S_{t+1}) | S_t = s] \\ &= \sum_a \pi(a|s) \sum_{s', r} (r + \gamma v_{\pi}(s')) p(s', r | s, a) \end{aligned}$$

Bellman Equation Using the MDP

- System of linear equations; one for each state s

$$\begin{aligned} v_{\pi}(s) &= E[R_{t+1} + \gamma v_{\pi}(S_{t+1}) | S_t = s] \\ &= \sum_a \pi(a|s) \sum_{s', r} (r + \gamma v_{\pi}(s')) p(s', r | s, a) \end{aligned}$$

- System of linear equations; one for each state (s, a)

$$\begin{aligned} q_{\pi}(s, a) &= E[R_{t+1} + \gamma v_{\pi}(S_{t+1}) | S_t = s, A_t = a] \\ &= \sum_{s', r} (r + \gamma v_{\pi}(s')) p(s', r | s, a) \end{aligned}$$

Optimal State and Action Value Functions

$$v_*(s) = \max_{\pi} v_{\pi}(s), s \in S$$

$$q_*(s, a) = \max_{\pi} q_{\pi}(s, a), \forall s, a$$

Optimal State and Action Value Functions

$V_*(s) \geq V_\pi(s)$ for all policies π .

$$v_*(s) = \max_{\pi} v_{\pi}(s), s \in S$$

$$q_*(s, a) = \max_{\pi} q_{\pi}(s, a), \forall s, a$$

Over the set of all policies (functions) that map states to actions, look for policies that for every state s in the set of states S , maximize the state's value.

Bellman Optimality Equations

We can write:

$$v_*(s) = E_{\pi^*} [R_{t+1} + \gamma v_*(S_{t+1}) | S_t = s]$$

Optimality principle:

- It is sufficient to pick the optimal action at time t (given $v_*(S_{t+1})$)

$$\rightarrow v_*(s) = \max_a E [R_{t+1} + \gamma v_*(S_{t+1}) | S_t = s, A_t = a]$$

$$\max_a q_*(s, a)$$

$S = \{s_1, s_2\}$
 $\pi_*^{(1)} \quad \pi_*^{(2)}$
 $V_{\pi_*^{(1)}}(s_1) > V_{\pi_*^{(2)}}(s_1)$
 $V_{\pi_*^{(1)}}(s_2) < V_{\pi_*^{(2)}}(s_2)$
 Neither policy is optimal. You could improve $\pi_*^{(1)}$ by setting $\pi_*^{(1)}(s_2) = \pi_*^{(2)}(s_2)$. $\pi_*^{(1)}$ now results in state values at least as large as $\pi_*^{(2)}$.